

Question 1

P1

Manitoba postal codes consist of three letters and three digits. Determine the total number of possible postal codes in Manitoba if the following characteristics must be met:

- The postal code must begin with the letter R.
- Only 18 letters of the alphabet can be used.
- The letters and digits must alternate.
- The letters and digits can be repeated.

Solution

$$\frac{1 \cdot 10 \cdot 18 \cdot 10 \cdot 18 \cdot 10}{R} = 324\,000$$

1 mark

Note:

The restriction of the first letter does not need to be shown.

Question 2

R10

The percent of people who click on a link, in response to seeing an advertisement on YouTube, can be modeled by the formula:

$$R(t) = -(0.8)^{0.2t} + 0.6$$

where $R(t)$ is the percentage, in decimal form, of people who click on the link

t is the time in days.

Determine, algebraically, the number of days it will take for 45% of people who see the advertisement to click on the link.

Solution

$$0.45 = -(0.8)^{0.2t} + 0.6$$

$$-0.15 = -(0.8)^{0.2t}$$

$$0.15 = (0.8)^{0.2t}$$

$$\log(0.15) = \log(0.8)^{0.2t}$$

$$\log(0.15) = 0.2t \cdot \log(0.8)$$

$$\frac{\log(0.15)}{0.2 \cdot \log(0.8)} = t$$

$$t = 42.508958\dots$$

$$t = 42.509 \text{ days}$$

½ mark for applying logarithms

1 mark for power law

½ mark for evaluating quotient of logarithms

2 marks

Question 3

P4

Determine and simplify the middle term in the binomial expansion of $\left(\frac{5}{x} + 4x^3\right)^6$.

Solution

$$t_4 = {}_6C_3 \left(\frac{5}{x}\right)^3 (4x^3)^3 \quad 2 \text{ marks (1 mark for } {}_6C_3; \frac{1}{2} \text{ mark for each consistent factor)}$$

$$= 20 \left(\frac{125}{x^3}\right) (64x^9)$$

$$= 160\,000x^6 \quad 1 \text{ mark for simplification (}\frac{1}{2} \text{ mark for coefficient; } \frac{1}{2} \text{ mark for exponent)}$$

3 marks

Question 4

T5

Solve $3\tan^2 x + 5\tan x - 6 = 0$, over the interval $[0, 2\pi]$.

Solution

$$\tan x = \frac{-5 \pm \sqrt{5^2 - 4(3)(-6)}}{2(3)}$$

$$\tan x = \frac{-5 \pm \sqrt{97}}{6}$$

1 mark for solving for $\tan x$

$$\tan x = 0.808\,142\dots$$

$$\tan x = -2.474\,809\dots$$

$$x_r = 0.679\,686\dots$$

$$x_r = 1.186\,784\dots$$

$$x = 0.680, 3.821$$

$$x = 1.955, 5.096$$

2 marks for solving for x ($\frac{1}{2}$ mark for each value)

3 marks

Question 5

T1

Given $\theta = -80^\circ$,

a) determine the measure of θ , in radians.

b) state all the coterminal angles of θ .

Solution

a) $\theta = -80^\circ \left(\frac{\pi}{180^\circ} \right)$

1 mark

$$\theta = -\frac{4\pi}{9}$$

or

$$\theta = -1.396$$

b) $\theta = -80^\circ + 360^\circ k, k \in \mathbb{Z}$

or

$$\theta = -\frac{4\pi}{9} + 2k\pi, k \in \mathbb{Z}$$

or

$$\theta = -1.396 + 2k\pi, k \in \mathbb{Z}$$

1 mark

Question 6

R12

State a possible value of n if the polynomial function $p(x) = (x+1)^n(x-4)^2$ has a range of $(-\infty, \infty)$.

Solution

$$n = 1$$

or

$$n = 3$$

1 mark

Note:

Accept any odd whole number for n .

Question 7

R2, R3, R5

Describe the transformations used to obtain the graph of the function $y = -2f(x + 8) - 5$ from the graph of $y = f(x)$.

Solution

The graph of $y = f(x)$ is reflected over the x -axis, vertically stretched by a factor of 2, translated left 8 units and down 5 units.

- 1 mark for vertical reflection
- 1 mark for vertical stretch
- 1 mark for horizontal translation
- 1 mark for vertical translation

4 marks

Note:

Deduct a maximum of 1 mark for concept error of incorrect order of vertical transformations.

Question 8

R14

State the domain and range of the function $f(x) = \frac{-2}{x^2}$.

Solution

Domain: $\{x|x \neq 0, x \in \mathbb{R}\}$ or $(-\infty, 0) \cup (0, \infty)$

1 mark for domain

Range: $\{y|y < 0, y \in \mathbb{R}\}$ or $(-\infty, 0)$

1 mark for range

2 marks

Question 9

P4

Given the following row of Pascal's Triangle, state the values in the next row.

1 5 10 10 5 1

Solution

1 6 15 20 15 6 1

1 mark

Question 10

R6

Given $f(x) = 3 - 2x$, determine the equation of $f^{-1}(x)$.

Solution

$$\begin{aligned}\text{Let } y &= f(x) \\ y &= 3 - 2x\end{aligned}$$

To determine the inverse of $f(x)$, switch x and y .

$$\begin{aligned}x &= 3 - 2y && \text{1 mark for switching } x \text{ and } y \\ 2y &= 3 - x \\ y &= \frac{3 - x}{2} && \frac{1}{2} \text{ mark for solving for } y \\ f^{-1}(x) &= \frac{3 - x}{2} && \frac{1}{2} \text{ mark for writing equation in terms of } f^{-1}(x)\end{aligned}$$

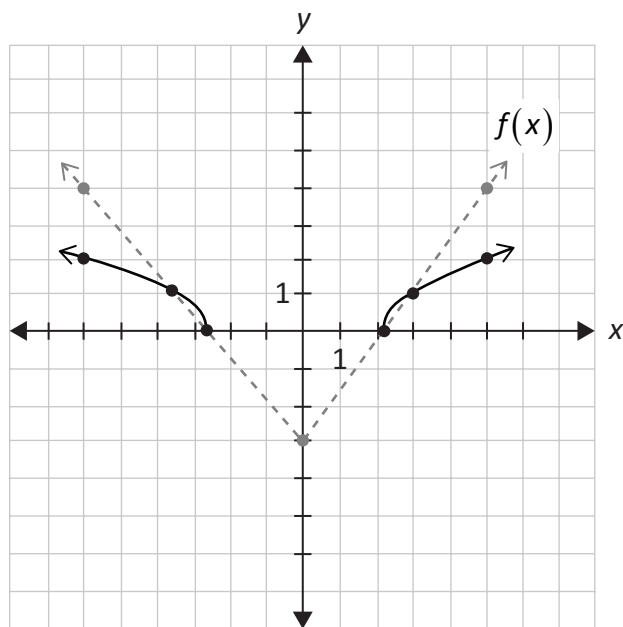
2 marks

Question 11

R13

Given the graph of $y = f(x)$, sketch the graph of $y = \sqrt{f(x)}$.

Solution



1 mark for restricting domain

$\frac{1}{2}$ mark for shape between invariant points, $\{0 \leq y \leq 1\}$

$\frac{1}{2}$ mark for shape above invariant points, $\{y \geq 1\}$

2 marks

Question 12

T6

Prove the identity for all permissible values of x .

$$\frac{\csc^2 x + \sec^2 x}{\tan x + \cot x} = \csc x \sec x$$

Solution

Method 1

Left-Hand Side	Right-Hand Side
$\frac{\frac{1}{\sin^2 x} + \frac{1}{\cos^2 x}}{\frac{\sin x}{\cos x} + \frac{\cos x}{\sin x}}$	$\csc x \sec x$
$\frac{\frac{\cos^2 x + \sin^2 x}{\sin^2 x \cos^2 x}}{\frac{\cos^2 x + \sin^2 x}{\cos x \sin x}}$	
$\frac{\cancel{\cos x} \cancel{\sin x}}{\sin^{\cancel{2}} x \cos^{\cancel{2}} x}$	
$\frac{1}{\sin x \cos x}$	
$\csc x \sec x$	

1 mark for correct substitution of identities

1 mark for algebraic strategies

1 mark for logical process to prove the identity

3 marks

Question 12

T6

Method 2

Left-Hand Side	Right-Hand Side
$\frac{1}{\sin^2 x} + \frac{1}{\cos^2 x}$ $\frac{\tan x}{1} + \frac{1}{\tan x}$ $\frac{\cos^2 x + \sin^2 x}{\sin^2 x \cos^2 x}$ $\frac{\tan^2 x + 1}{\tan x}$ $\frac{1}{\sin^2 x \cos^2 x}$ $\frac{\sec^2 x}{\tan x}$ $\frac{\tan x}{\sin^2 x \cancel{\cos^2 x} \cancel{\sec^2 x}}$ $\frac{\cancel{\sin x}}{\cos x}$ $\frac{\sin^2 x}{1}$ $\frac{1}{\sin x \cos x}$ $\csc x \sec x$	$\csc x \sec x$

1 mark for correct substitution of identities

1 mark for algebraic strategies

1 mark for logical process to prove the identity

3 marks

Question 13

R9

Justify that the graph of $f(x) = e^{x+1} + 5$ does not have an x-intercept.

Solution

$$0 = e^{x+1} + 5$$

$$-5 = e^{x+1}$$

$e^{x+1} > 0$, therefore there is no solution.

or

The range of the graph is $\{y \mid y > 5, y \in \mathbb{R}\}$.

or

The horizontal asymptote is $y = 5$ and it is an increasing exponential function above the asymptote.

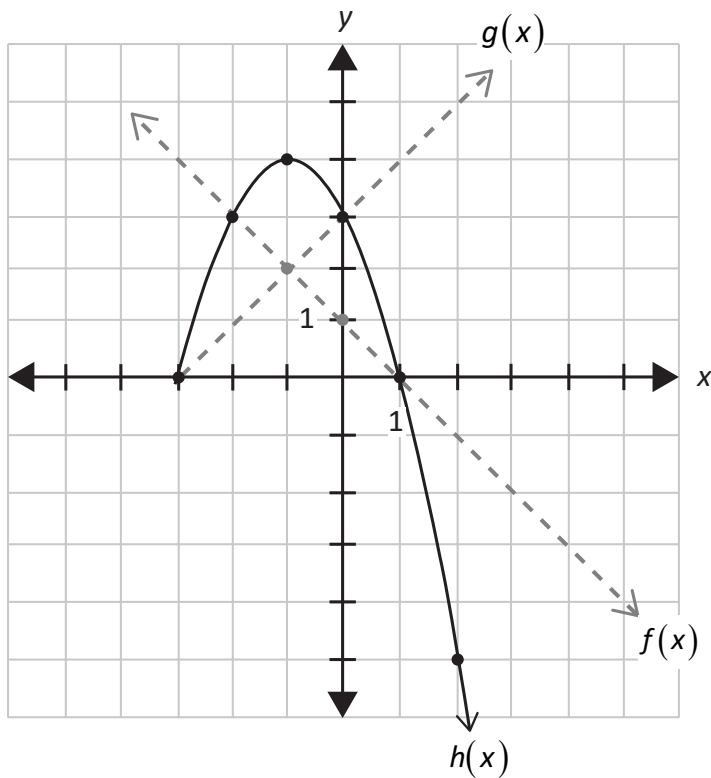
1 mark

Question 14

R1

Given the graphs of $f(x)$ and $g(x)$, sketch the graph of $h(x) = f(x) \cdot g(x)$.

Solution



1 mark for operation of multiplication
1 mark for restricted domain

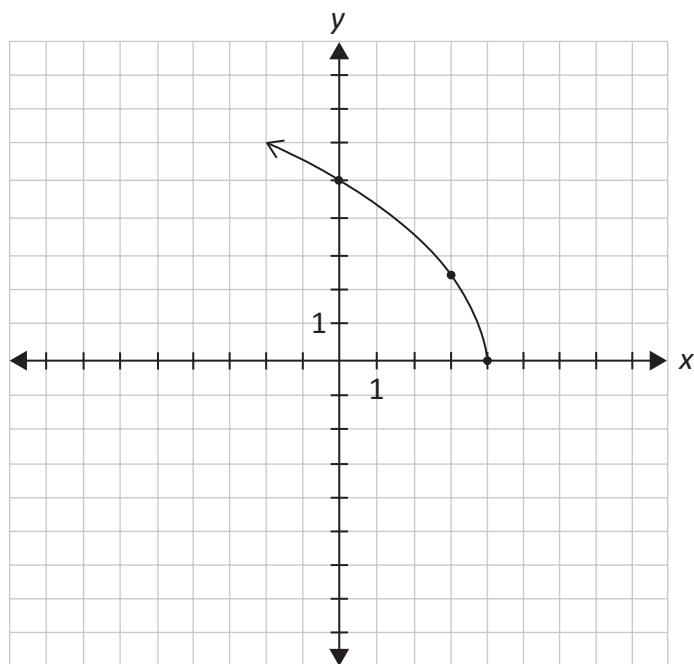
2 marks

Question 15

R13

Sketch the graph of $f(x) = \frac{5}{2}\sqrt{-(x-4)}$.

Solution



- 1 mark for shape of a radical function
- 1 mark for vertical stretch
- 1 mark for horizontal reflection
- 1 mark for horizontal translation

4 marks

Question 16**R11**

Identify the number of x -intercepts on the graph of the polynomial function,

$$p(x) = 3x(x - 8)(x^2 + 5).$$

a. 1

b. 2

c. 3

d. 4

Question 17**R5**

Identify the equation that represents the graph of $y = x^2 - 3$ after a reflection over the x -axis.

a. $y = -x^2 - 3$

b. $y = x^2 - 3$

c. $y = x^2 + 3$

d. $y = -x^2 + 3$

Question 18**R7**

Given $2^{\log_2 3} = x$, identify the value of x .

a. 1

b. 3

c. 6

d. 8

Question 19

R14

Identify the y-intercept on the graph of $y = \frac{x-3}{x^2-5x+6}$.

a. -3

b. -2

c. $-\frac{1}{2}$

d. 3

Question 20

P2

Identify the total number of arrangements for 10 students and 2 teachers to sit in a row if the teachers must sit together.

a. $10!2!$

b. $11!$

c. $11!2!$

d. $12!$

Question 21

R11

Identify which expression represents the remainder when the polynomial $P(x)$ is divided by $(x-4)$.

a. $P(-4)$

b. $P(4)$

c. $P(x-4)$

d. $P(x+4)$

Question 22

T4

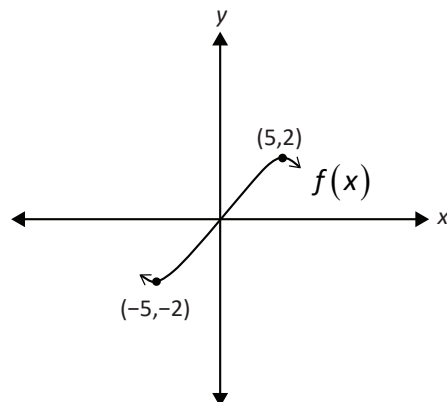
Identify the period of the graph of the sinusoidal function.

a. 10π

b. 10

c. 20π

d. 20



Question 23

R8

Identify an equivalent expression for $\log_3 9 + \log_3 5$.

a. $\log_3 14$

b. $2\log_3 5$

c. $\log_3 45$

d. $\log_3 \left(\frac{9}{5} \right)$

Question 24

T1

Identify the coterminal angle(s) of $\theta = -\frac{11\pi}{10}$ over the interval $[-3\pi, \pi]$.

a. $\theta = \frac{9\pi}{10}$

b. $\theta = -\frac{31\pi}{10}, -\frac{21\pi}{10}$

c. $\theta = -\frac{31\pi}{10}, \frac{9\pi}{10}$

d. $\theta = -\frac{21\pi}{10}, \frac{9\pi}{10}$

Question 25

R11

Express $p(x) = x^3 - 13x - 12$ in completely factored form.

Solution

$$p(-1) = (-1)^3 - 13(-1) - 12$$

$$p(-1) = 0$$

$\therefore (x + 1)$ is a factor

1 mark for identifying one possible zero of $P(x)$

-1	1	0	-13	-12
		-1	1	12
	1	-1	-12	0

1 mark for synthetic division (or equivalent strategy)

$$p(x) = (x + 1)(x^2 - x - 12)$$

$$p(x) = \underline{(x + 1)(x - 4)(x + 3)}$$

1 mark for consistent product of factors

3 marks

Question 26

T5

Verify, by substitution, that the equation $\frac{\cos^2 \theta}{\sin^2 \theta} = \frac{2 \csc \theta}{\sec^2 \theta}$ is true for $\theta = \frac{\pi}{6}$.

Solution

Left-Hand Side	Right-Hand Side	
$\frac{\cos^2\left(\frac{\pi}{6}\right)}{\sin^2\left(\frac{\pi}{6}\right)}$	$\frac{2 \csc\left(\frac{\pi}{6}\right)}{\sec^2\left(\frac{\pi}{6}\right)}$	
$\frac{\left(\frac{\sqrt{3}}{2}\right)^2}{\left(\frac{1}{2}\right)^2}$	$\frac{2(2)}{\left(\frac{2}{\sqrt{3}}\right)^2}$	2 marks for exact values (½ mark for each)
$\frac{\frac{3}{4}}{\frac{1}{4}}$	$\frac{2(2)}{\left(\frac{4}{3}\right)}$	
	$\frac{12}{4}$	
3	3	1 mark for simplification (½ mark for LHS; ½ mark for RHS)

3 marks

Question 27

R9

Given $f(x) = \log_3 x + 2$,

a) determine the x-intercept of $f(x) = \log_3 x + 2$.

b) sketch the graph of $f(x) = \log_3 x + 2$.

Solution

a)

$$0 = \log_3 x + 2$$

$$-2 = \log_3 x$$

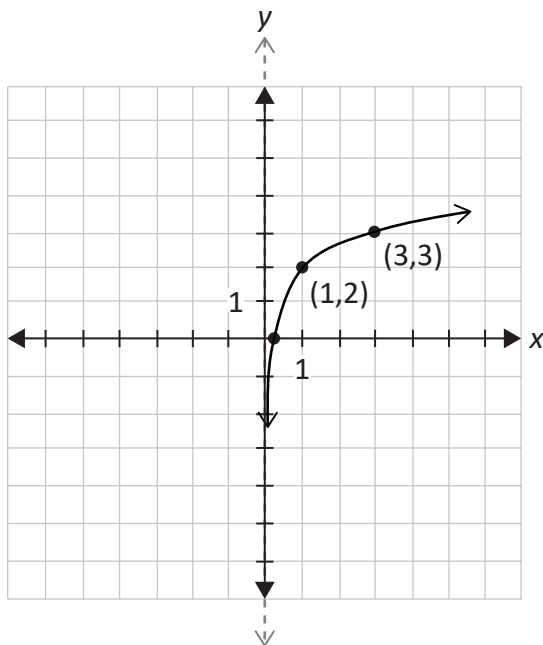
$$3^{-2} = x$$

$$\frac{1}{3^2} = x$$

$$\frac{1}{9} = x$$

1 mark

b)



1 mark for shape of a logarithmic function
1 mark for vertical translation

2 marks

Question 28

T3

Evaluate.

$$\sec^2\left(\frac{5\pi}{4}\right) \cdot \tan^2\left(-\frac{2\pi}{3}\right)$$

Solution

$$\left(-\frac{2}{\sqrt{2}}\right)^2 \left(\frac{\sqrt{3}}{1}\right)^2$$

1 mark for $\sec\left(\frac{5\pi}{4}\right)$ (½ mark for quadrant; ½ mark for value)

$$\left(\frac{4}{2}\right)\left(\frac{3}{1}\right)$$

1 mark for $\tan\left(-\frac{2\pi}{3}\right)$ (½ mark for quadrant; ½ mark for value)

6

2 marks

Question 29

R1

Given the functions $f(x) = x - 3$ and $g(x) = \sqrt{x} + 2$, state the domain of $g(f(x))$.

Solution

$$\{x \mid x \geq 3, x \in \mathbb{R}\}$$

1 mark

or

$$[3, \infty)$$

Question 30

T6

Evaluate.

$$\sin 70^\circ \cos 25^\circ - \cos 70^\circ \sin 25^\circ$$

Solution

$$\sin 70^\circ \cos 25^\circ - \cos 70^\circ \sin 25^\circ = \sin(70^\circ - 25^\circ) \quad \frac{1}{2} \text{ mark for substitution into correct identity}$$

$$= \sin 45^\circ$$

$$= \frac{\sqrt{2}}{2}$$

$\frac{1}{2}$ mark for exact value

1 mark

Question 31

R10

Solve, algebraically.

$$\log_{15}(x^2 - 1) = 1$$

Solution

Method 1

$$15^1 = x^2 - 1$$

$$0 = x^2 - 16$$

$$0 = (x + 4)(x - 4)$$

$$x = -4, x = 4$$

1 mark for exponential form

1 mark for solving for x ($\frac{1}{2}$ mark for each value)

2 marks

Method 2

$$\log_{15}(x^2 - 1) = \log_{15} 15$$

$$x^2 - 1 = 15$$

$$x^2 = 16$$

$$x = \pm 4$$

$\frac{1}{2}$ mark for logarithmic form

$\frac{1}{2}$ mark for equating arguments

1 mark for solving for x ($\frac{1}{2}$ mark for each value)

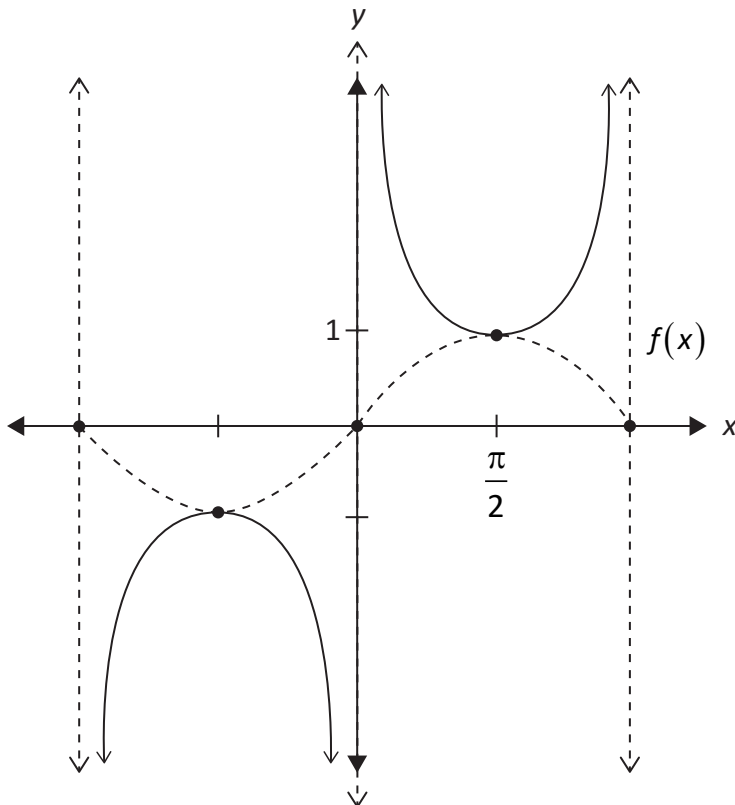
2 marks

Question 32

R1

Given the graph of $y = f(x)$, sketch the graph of $y = \frac{1}{f(x)}$.

Solution



1 mark for asymptotic behaviour approaching $x = -\pi$, $x = 0$, and $x = \pi$

1 mark for invariant points ($\frac{1}{2}$ mark for $y = -1$; $\frac{1}{2}$ mark for $y = 1$)

2 marks

Question 33

T4

A light is attached to the end of a blade of a wind turbine. The height of the light, with respect to the ground, follows the sinusoidal equation:

$$h(t) = 80 \cos\left(\frac{\pi}{2}t\right) + 240$$

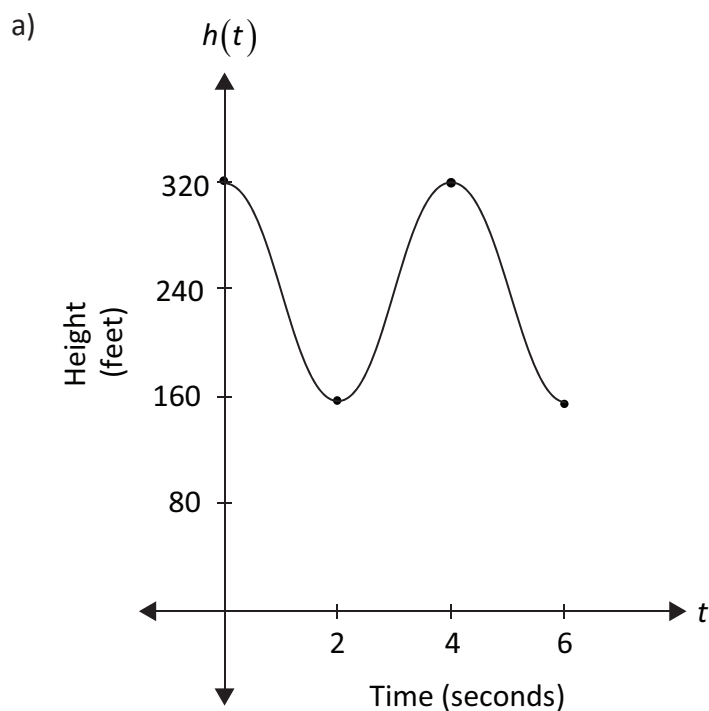
where $h(t)$ is the height of the light with respect to the ground, measured in feet and,

t is the time, measured in seconds.



- Sketch the graph of the height of the light with respect to the ground, over an interval of 6 seconds.
- If the turbine begins to rotate faster, describe the resulting effect on the period of the graph of $h(t)$.

Solution



1 mark for shape of a sinusoidal function with correct amplitude
1 mark for period
1 mark for vertical translation

3 marks

1 mark

- The period would decrease.

Question 34

P2

Elodie was asked to determine the total number of arrangements of the letters in the word EXCELLENCE.

Her solution:

$$10! = 3628\ 800$$

Describe her error.

Solution

She did not consider the repeating letters.

1 mark

Question 35

T6

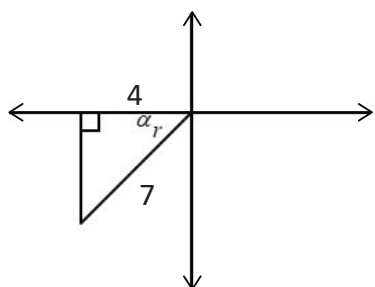
Given that $\cos \alpha = -\frac{4}{7}$ where α is in quadrant III and $\sin \beta = \frac{5}{13}$ where β is in quadrant II, determine the exact value of:

a) $\cos(\alpha + \beta)$

b) $\sec(\alpha + \beta)$

Solution

a)

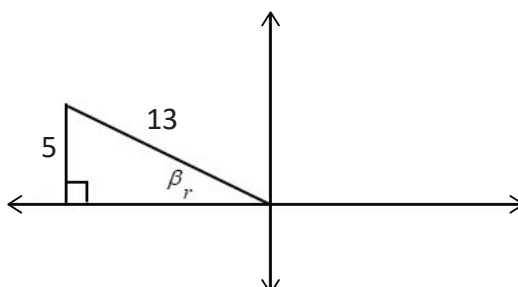


$$x^2 + y^2 = r^2$$

$$16 + y^2 = 49$$

$$y^2 = 33$$

$$y = \pm\sqrt{33}$$



$$x^2 + y^2 = r^2$$

$$x^2 + 25 = 169$$

$$x^2 = 144$$

$$x = \pm 12$$

$\frac{1}{2}$ mark for value of x

$\frac{1}{2}$ mark for value of y

$$\cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta$$

$$= \left(-\frac{4}{7}\right)\left(-\frac{12}{13}\right) - \left(-\frac{\sqrt{33}}{7}\right)\left(\frac{5}{13}\right)$$

$$= \frac{48 + 5\sqrt{33}}{91}$$

$\frac{1}{2}$ mark for consistent value of $\cos \beta$

$\frac{1}{2}$ mark for consistent value of $\sin \alpha$

1 mark for substitution into correct identity

3 marks

b) $\sec(\alpha + \beta) = \frac{91}{48 + 5\sqrt{33}}$

1 mark for consistent value of $\sec(\alpha + \beta)$

1 mark

Note:

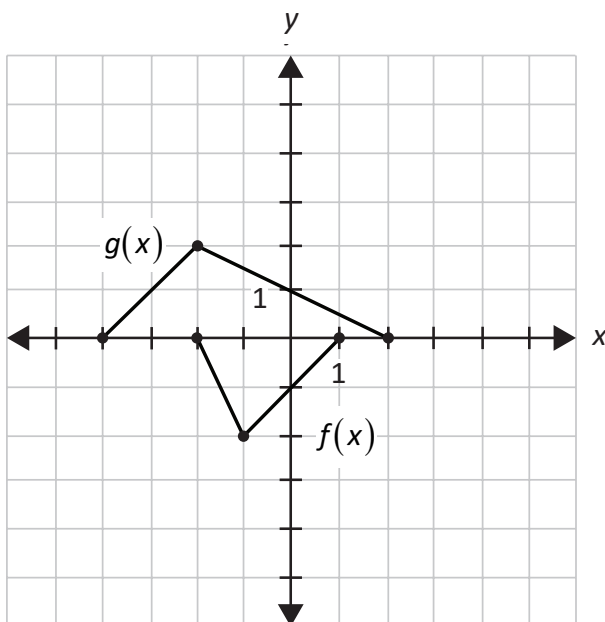
Accept any of the following values for x : $x = \pm 12$, $x = -12$, or $x = 12$.

Accept any of the following values for y : $y = \pm\sqrt{33}$, $y = -\sqrt{33}$, or $y = \sqrt{33}$.

Question 36

R4, R5

Determine the equation for $g(x)$, in terms of $f(x)$.



Solution

$$g(x) = -f\left(\frac{x}{2}\right)$$

1 mark for vertical reflection
1 mark for horizontal stretch

2 marks

Question 37

R7

State possible values for m and n that satisfy the equation, $\log_m n = 5$.

Solution

Some possible solutions are:

$$m = 2 \quad n = 32$$

or

$$m = 3 \quad n = 243$$

or

$$m = 4 \quad n = 1024$$

or

$$m = 5 \quad n = 3125$$

1 mark

Notes:

Other values for m and n are possible.

Question 38

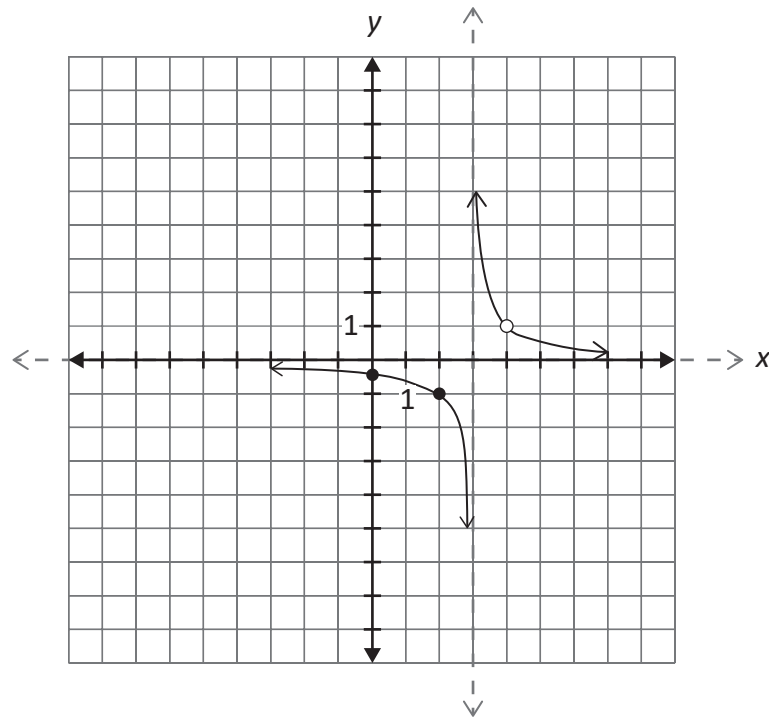
R14

Sketch the graph of $f(x) = \frac{x-4}{x^2-7x+12}$.

Solution

$$f(x) = \frac{\cancel{(x-4)}}{\cancel{(x-4)}(x-3)} \\ = \frac{1}{x-3}, \quad x \neq 4$$

$\therefore$ there is a point of discontinuity (hole) at (4,1)



1 mark for asymptotic behaviour approaching $x = 3$

1 mark for asymptotic behaviour approaching $y = 0$

1 mark for point of discontinuity (hole) at $x = 4$

1 mark for shape of rational function ($\frac{1}{2}$ mark for graph left of $x = 3$, $\frac{1}{2}$ mark for graph right of $x = 3$)

4 marks

Notes:

Deduct $\frac{1}{2}$ mark for procedural error (incorrect y-value for point of discontinuity (hole)).

Question 39

P2

Justify that the value of ${}_3P_4$ does not exist.

Solution

Method 1

When using ${}_nP_r$, $n \geq r$.

Method 2

$${}_3P_4 = \frac{3!}{(3-4)!}$$

$${}_3P_4 = \frac{3!}{(-1)!}$$

$(-1)!$ does not exist, therefore ${}_3P_4$ does not exist.

1 mark

Question 40

T4

State the range of the sinusoidal function that has the following characteristics:

- an amplitude of 3
- a maximum at $\left(\frac{\pi}{2}, 1\right)$

Solution

$$\{y \mid -5 \leq y \leq 1, y \in \mathbb{R}\}$$

or

$$[-5, 1]$$

1 mark

Question 41

P3

Solve, algebraically.

$${}_nC_2 = 15$$

Solution

$$\frac{n!}{2!(n-2)!} = 15$$

½ mark for substitution

$$\frac{n(n-1)\cancel{(n-2)!}}{2!\cancel{(n-2)!}} = 15$$

1 mark for factorial expansion

½ mark for simplification of factorials

$$n(n-1) = 30$$

$$n^2 - n - 30 = 0$$

$$(n-6)(n+5) = 0$$

$$n = 6 \quad \cancel{n = -5}$$

½ mark for the permissible value of n

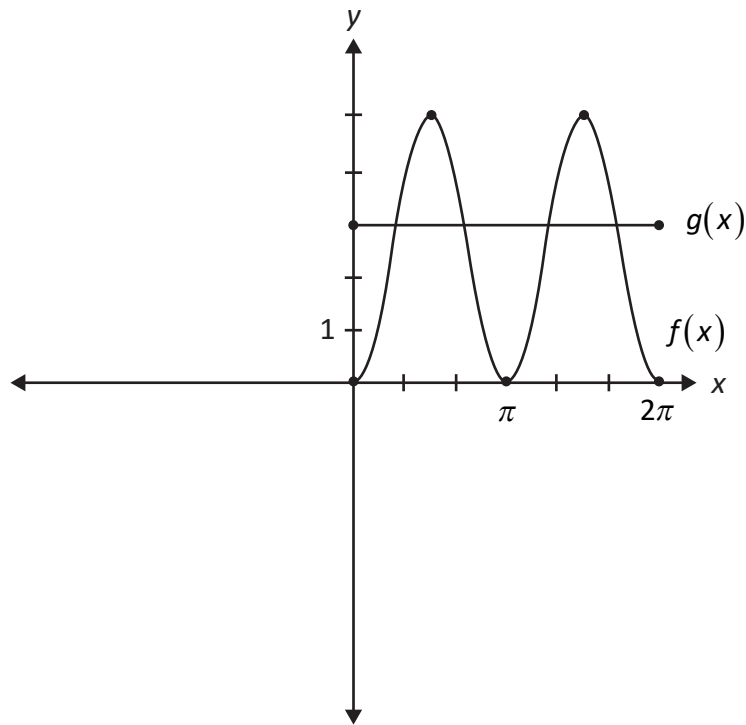
½ mark for showing the rejection of the extraneous root

3 marks

Question 42

T5

Describe how to use the graphs of $f(x) = 5\sin^2 x$ and $g(x) = 3$ to solve the equation, $5\sin^2 x = 3$, over the interval $[0, 2\pi]$.



Solution

The solution will be the x -values where the two graphs intersect.

1 mark

Question 43

R8

Solve, algebraically.

$$2\log_3 5 - \frac{1}{3}\log_3 125 = \log_3 a$$

Solution

Method 1

$$\log_3 5^2 - \log_3 125^{\frac{1}{3}} = \log_3 a$$

1 mark for power law ($\frac{1}{2}$ mark for each)

$$\log_3 25 - \log_3 5 = \log_3 a$$

$$\log_3 \left(\frac{25}{5} \right) = \log_3 a$$

1 mark for quotient law

$$\log_3 (5) = \log_3 a$$

$$a = 5$$

1 mark for equating arguments

3 marks

Method 2

$$\log_3 5^2 - \log_3 125^{\frac{1}{3}} - \log_3 a = 0$$

1 mark for power law ($\frac{1}{2}$ mark for each)

$$\log_3 \left(\frac{25}{5a} \right) = 0$$

1 mark for quotient law

$$3^0 = \frac{25}{5a}$$

1 mark for exponential form

$$5a = 25$$

$$a = 5$$

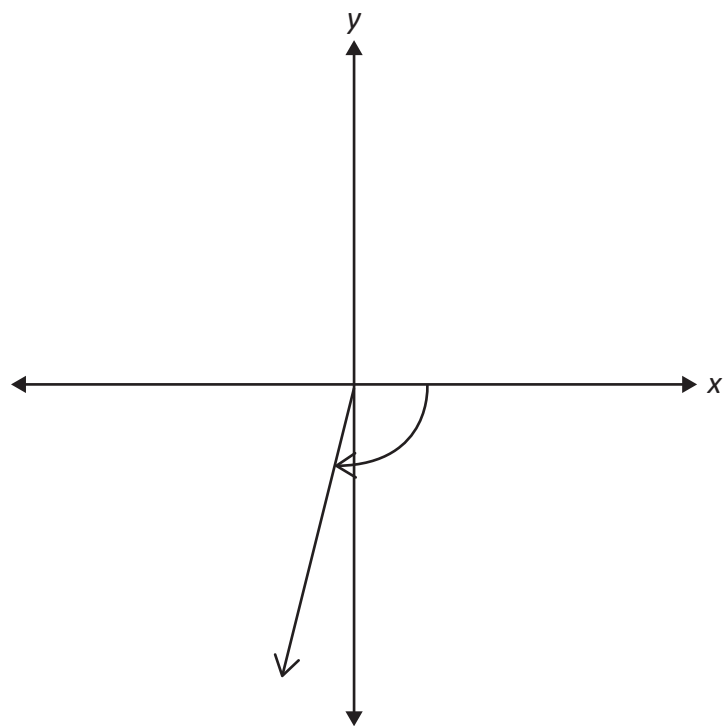
3 marks

Question 44

T1

Sketch the angle, $\frac{-7\pi}{12}$, in standard position.

Solution



½ mark for an appropriate angle
in quadrant III
½ mark for correct direction

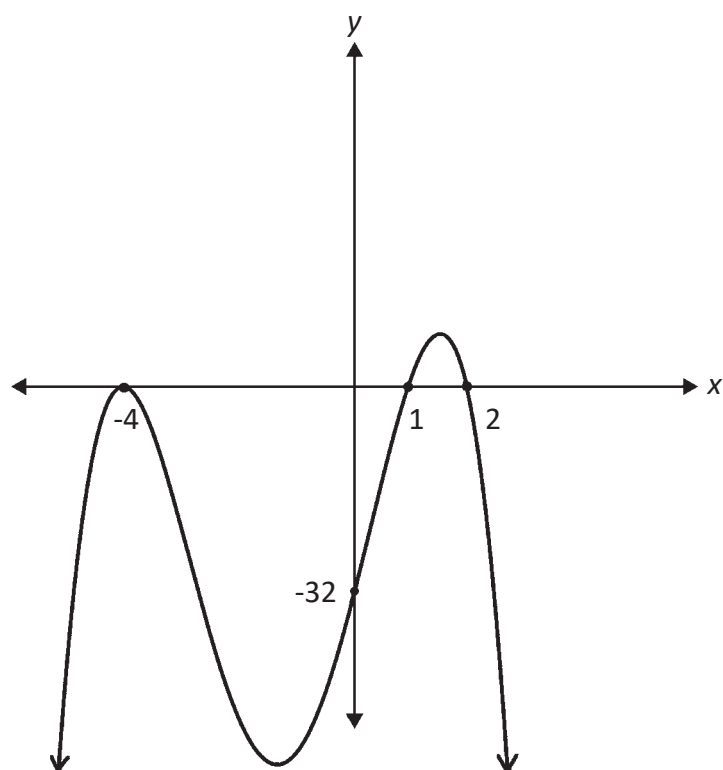
1 mark

Question 45

R12

Sketch the graph of $p(x) = -(x-1)(x-2)(x+4)^2$.

Solution

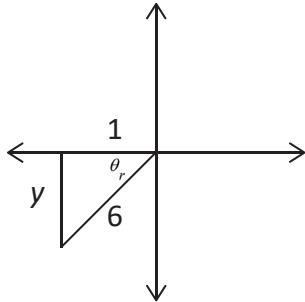


1 mark for x-intercepts
½ mark for y-intercept
1 mark for multiplicity of 2 at $x = -4$
½ mark for end behaviour

3 marks

Question 46**T3**

The point, $P(\theta) = \left(-\frac{1}{6}, y\right)$, lies on the unit circle and is located in Quadrant III. Determine the exact value of $\csc \theta$.

Solution

$$(-1)^2 + y^2 = (6)^2$$

½ mark for substitution

$$1 + y^2 = 36$$

$$y^2 = 35$$

$$y = \pm\sqrt{35}$$

½ mark for solving for y

$$\csc \theta = -\frac{6}{\sqrt{35}}$$

1 mark for consistent value of $\csc \theta$

(½ mark for quadrant; ½ mark for value)

or

2 marks

$$\csc \theta = -\frac{6\sqrt{35}}{35}$$

Notes:

Accept any of the following values for y : $y = \pm\sqrt{35}$, $y = \sqrt{35}$, or $y = -\sqrt{35}$.